# Coercive Sesquilinear Forms

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Our aim in this text is to present the general theory of coercive sesquilinear forms and its applications to stationary and evolutional linear equations. Sesquilinear forms are mostly efficient when studying abstract problems in the Hilbert space setting.

## 1. Generalities

By a *sesquilinear form* defined in complex linear space $V$ is meant a mapping $\mathfrak{a}: V \times V \to \mathbb{C}$, which is linear in the first and conjugate linear in the second argument, that is, for all $x, y, z \in V$ and $\lambda, \mu \in \mathbb{C}$ the computation rules

$$\begin{cases} \mathfrak{a}(x+y,z) \;=\; \mathfrak{a}(x,z) \;+\; \mathfrak{a}(y,z) \\ \mathfrak{a}(x,y+z) \;=\; \mathfrak{a}(x,y) \;+\; \mathfrak{a}(x,z) \\ \mathfrak{a}(\lambda x, \mu y) \;=\; \lambda\overline{\mu}\,\mathfrak{a}(x,y) \end{cases}$$

are valid. If the linear space $V$ is real, however, then $\mathfrak{a}$ is assumed to be linear also in the second argument, so it becomes into a *bilinear form* in this case.

The mapping $x \in V \mapsto \mathfrak{a}[x] := \mathfrak{a}(x,x) \in \mathbb{C}$ is called the *quadratic form* of $\mathfrak{a}$. Notice that all values $\mathfrak{a}(x,y)$ can be recovered from the values $\mathfrak{a}[x]$ and $\mathfrak{a}[y]$ by certain polarization formulas.

A sesquilinear form $\mathfrak{a}$ may also be regarded as a linear mapping $\mathcal{A}_\mathfrak{a}: V \to V_a^\#$, where $V_a^\#$ is the (algebraic) conjugate space of $V$, that is, the linear space of conjugate linear functionals $\phi: V \to \mathbb{C}$. More precisely, given a sesquilinear form $\mathfrak{a}$, each element $x \in V$ gives rise to a conjugate linear functional $\phi_x: V \to \mathbb{C}$, defined by $\phi_x(y) := \mathfrak{a}(x,y)$ for $y \in V$. Let $V_a^\#$ again be the linear space of conjugate linear functionals on $V$. Then the mapping

$$\mathcal{A}_\mathfrak{a}: \begin{cases} \mathcal{A}_\mathfrak{a}: V \to V^\#, \\ \mathcal{A}_\mathfrak{a}\, x := \phi_x \quad (x \in V) \end{cases}$$

proves to be linear and is said to be the *form operator* or the *representation operator* associated with $\mathfrak{a}$.

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Reversely, for any linear operator $\mathcal{A}: V \to V_a^{\#}$, there is a sesquilinear form $\mathfrak{a}: V \times V \to \mathbb{K}$ such that $\mathfrak{a}(x,y) = (\mathcal{A}x)y$ for $x, y \in V$. Hence sesquilinear forms acting on a complex linear space $V$ appear in a one-to-one correspondence to linear mappings acting between the spaces $V$ and $V_a^{\#}$. This correspondence remains valid, if $V$ is a *real* linear space, $\mathfrak{a}$ is a *bilinear* form and $\mathcal{A}$ is the associated mapping from $V$ to the adjoint space $V^{\#}$.

## 2. Continuous and Coercive Forms

From now on we assume that $V$ is a *normed* linear space with underlying field $\mathbb{C}$. Then any sesquilinear form $\mathfrak{a}: V \times V \to \mathbb{C}$ is called *bounded*, if there is $\Theta > 0$ such that

$$|\mathfrak{a}(x,y)| \leq \Theta \, \|x\| \cdot \|y\| \qquad (x, y \in V).$$

Due to the *uniform boundedness principle*, this property is equivalent to say that $\mathfrak{a}$ is continuous on the space $V \times V$.

The prototypical example of a continuous form is the inner product in a complex Hilbert space $H$, where boundedness is ensured by the CAUCHY-SCHWARZ inequality $|\langle x, y\rangle| \leq \|x\| \cdot \|y\|$.

Moreover, for each continuous sesquilinear form $\mathfrak{a}$ on $H$ there is a bounded linear operator $B_{\mathfrak{a}}: H \to H$ such that

$$\mathfrak{a}(x,y) \;=\; \langle B_{\mathfrak{a}} x, y\rangle \quad \text{ for all } x, y \in H, \tag{2.1}$$

which is a representation for the form $\mathfrak{a}$. In fact, if $x \in H$ is fixed, then the mapping $\mathfrak{a}(x,\cdot): H \to \mathbb{C}$ causes a continuous antilinear functional $\phi_x \in H_a^*$, and by RIESZ's representation theorem there is $\tilde{x} \in H$ such that $\phi_x(y) = \langle \tilde{x}, y\rangle$ for all $y \in H$. This suggests to define $B_{\mathfrak{a}} x := \tilde{x}$ for every $x \in H$. If (2.1) holds for some operator $B \in \mathscr{L}(H)$, then the form $\mathfrak{a}$ is said to be represented by $B$.

Finally, any continuous sesquilinear form $\mathfrak{a}: H \times H \to \mathbb{C}$ is called *coercive* if there is $\theta > 0$ such that

$$\operatorname{Re} \mathfrak{a}(x,x) \geq \theta \, \|x\|^2 \quad \text{ for all } x \in H. \tag{2.2}$$

The estimate (2.2) especially implies the validness of

$$\|B_{\mathfrak{a}} x\| \, \|x\| \;\geq\; |\langle B_{\mathfrak{a}} x, x\rangle| = |\mathfrak{a}(x,x)| \;\geq\; \operatorname{Re} \mathfrak{a}(x,x) \;\geq\; \theta \, \|x\|^2$$

for all $x \in H$. Hence the representation operator $B_{\mathfrak{a}}$ is bounded below with lower bound $\theta$, and we may conclude by the fundamental *bounded-below theorem* that it is injective and the range $\mathfrak{R}(B_{\mathfrak{a}})$ forms a closed linear subspace in $H$.

## 3. The Variational Approach

Coercivity of a sesquilinear form $\mathfrak{a}$ is of great interest with respect to the solvability of the so-called *variational problem*, which may be formulated as follows: given a continuous and conjugate linear functional $\phi \in V_a^*$, it is to find $x \in V$ such that

$$\mathfrak{a}(x, y) = \phi(y) \quad \text{for all } y \in V. \tag{3.1}$$

The LAX-MILGRAM *theorem* is of outstanding meaning in applied functional analysis and ensures well-posedness of the variational problem for any right-hand side $\phi \in V_a^*$:

THEOREM 3.1. If $\mathfrak{a} : V \times V \to \mathbb{C}$ is bounded and coercive, then for each $\phi \in V_a^*$ the equation (3.1) owns exactly one solution $x_\phi \in V$. In addition, $\| x_\phi \| \leq \theta^{-1} \|\phi\|$ holds.

In other words, if $\mathfrak{a}$ is bounded and coercive, then the form operator $\mathcal{A}_\mathfrak{a} : V \to V_a^*$ is bjective. The bounded inverse theorem implies that the inverse operator $\mathcal{A}_\mathfrak{a}{}^{-1}$ is bounded, too, so the solution $x_\phi$ of (3.1) depends continuously on $\phi$.

There is also an *operator version* of theorem 3.1. Let $A : V \to V$ be linear and bounded. We shall denote $A$ *coercive* if

$$\mathrm{Re}\langle Ax, x\rangle > \theta \|x\|^2$$

holds for some $\theta > 0$ and all $x \in V$. It should be clear that the mapping $(x, y) \mapsto \langle Ax, y\rangle$ defines a coercive form on $V$. Then the LAX-MILGRAM theorem implies invertibility of $A$ and boundedness of $A^{-1}$, where $\|A^{-1}\| \leq \theta^{-1}$ holds.

## 4. Operators Associated with H-Elliptic Forms

It is often indicated to study abstract linear problems in the setting of so-called *pairs* $(V, H)$ of Hilbert spaces, where the space $V$ is continuously and densely embedded into a second space $H$. More precisely, we assume that

(T-1) there is a linear mapping $\iota_1 : V \to H$ such that $\overline{\Re(\iota_1)} = H$;
(T-2) there is $c > 0$ such that $\|\iota_1 v\|_H \leq c \|v\|_V$ for all $v \in V$.

By means of the embedding $\iota_1$, one can also think of $V$ as a dense linear subspace in the space $H$.

Each Hilbert pair $(V, H)$ with embedding $\iota : V \to H$ gives rise to introduce the *transposed pair* $(H_a^*, V_a^*)$, where the conjugate space

$H_a^*$ can be seen as continuously and densely embedded into the conjugate space $V_a^*$ by a further embedding $\iota_2 : H_a^* \to V_a^*$. Then, by identifying the spaces $H$ and $H_a^*$ via the RIESZ map, combining both Hilbert pairs leads to the triple

$$V \overset{\iota_1}{\hookrightarrow} H \overset{\iota_2}{\hookrightarrow} V_a^*, \tag{4.1}$$

sometimes called a *Gelfand triple*, an *evolution triple* or a *rigged Hilbert space*. The center space $H$ is often denoted the *pivot space* of (4.1).

Gelfand triples own a remarkable property. Let $(\,\cdot\,,\,\cdot\,)$ and $\langle\,\cdot\,,\,\cdot\,\rangle$ be the inner products in $V$ and $H$, respectively, then for fixed chosen $x \in H$ the action of the antilinear linear functional $f_x := \iota_2\, x \in V_a^*$ to any element $v \in V$ is represented by the inner product of $x$ and $v$ in $H$, i. e. $f_x(v) = \langle x, \iota_1\, v\rangle$ holds for all $v \in V$.

The presented theory of coercive sesquilinear forms in a Hilbert space $V$ is often not sufficent for applications. Hence it is quite natural to examine sesquilinear forms also with respect to pairs $(V, H)$ of Hilbert spaces. In applications it is often the case that sesquilinear forms are merely continuous but not coercive. In the setting of a Hilbert pair $(V, H)$, however, we try to make a form $\mathfrak{a}$ coercive by adding an additional term, which is just a multiplicity of the inner product in $H$.

Indeed, if $\mathfrak{a} : V \times V \to \mathbb{C}$ is continuous and the slightly modified form

$$\mathfrak{a}_\lambda(u, v) := \mathfrak{a}(u, v) + \lambda\langle \iota_1 u, \iota_1 v\rangle \quad (u, v \in V)$$

is coercive for some $\lambda \in \mathbb{R}$, that is, there is $\theta > 0$ such that

$$\operatorname{Re}\mathfrak{a}(u, u) + \lambda\, \|\iota_1 u\|_H^2 \geq \theta\, \|u\|_V^2 \quad (u \in V), \tag{4.2}$$

then $\mathfrak{a}$ is called *quasi-coercive*, *$H$-elliptic* or simply a *Gårding form*. It should be clear that if $\mathfrak{a}_{\lambda_0}$ is coercive for some $\lambda_0 \in \mathbb{R}$, then $\mathfrak{a}_\lambda$ is so for $\lambda > \lambda_0$.

Quasi-coercive sesquilinear forms are often used to define densely defined and closed linear operators acting in the pivot space $H$ of a given Gelfand triple.

Let $\mathfrak{a}$ be a sesquilinear form defined on a Hilbert space $V$, which is the left hand side of a Gelfand triple $(V, H, V_a^*)$ with continuous embeddings $\iota_1 : V \to H$ and $\iota_2 : H \to V_a^*$. It is our aim to introduce a linear operator $A_\mathfrak{a}$ with domain space $H$ and range space $H$. This

will be done by restricting the form operator $\mathcal{A}_\mathfrak{a} : V \to V_a^*$ associated with $\mathfrak{a}$ to those elements $u \in V$, for which $\mathcal{A}_\mathfrak{a} u \in H$. Indeed, the domain $\mathfrak{D}(A_\mathfrak{a})$ is defined to be the subspace of those elements $u \in V$, for which $\mathcal{A}_\mathfrak{a} u \in \iota_2(H) \subset V_a^*$ holds, briefly denoted

$$\mathfrak{D}(A_\mathfrak{a}) := \{ u \in V : \mathcal{A}_\mathfrak{a} u \in H \}.$$

Clearly, the set $\mathfrak{D}(A_\mathfrak{a})$ can be regarded as a linear subspace of $H$ by means of the embedding $\iota_1$. Then for $u \in \mathfrak{D}(A)$ there exists $y \in H$ such that $\iota_2 y = \mathcal{A}_\mathfrak{a} u$. We set $x := \iota_1 u$ and $A_\mathfrak{a} x := y$ for $x \in \mathfrak{D}(A_\mathfrak{a})$ and denote $A_\mathfrak{a}$ the *operator in $H$ associated with* $(V, \mathfrak{a})$, or shortly, the *variational operator* derived from $\mathfrak{a}$. Another phrase is to say that the operator $A_\mathfrak{a}$ is the *part of* $\mathcal{A}_\mathfrak{a}$ *in* $H$.

The following theorem collects a couple of nice properties of the operator $A_\mathfrak{a}$ associated with a continuous and quasi-coercive form $\mathfrak{a}$. Besides the LAX-MILGRAM theorem, is is our second important theorem in this text:

THEOREM 4.1. Let $(V, H, V_a^*)$ be a triple of Hilbert spaces and the sesquilinear form $\mathfrak{a}$ be continuous and quasi-coercive on $V$ with parameter $\lambda_0 \in \mathbb{R}$. Then the following assertions are true:

(a) $A_\mathfrak{a}$ is closed.
(b) $A_\mathfrak{a}$ is densely defined in $H$.
(c) If the embedding of $V$ into $H$ is *compact*, then the operator $A_\mathfrak{a} + \lambda$ has a compact inverse $(A_\mathfrak{a} + \lambda)^{-1}$ for each $\lambda > \lambda_0$.
(d) The operator $A_\mathfrak{a} + \lambda$ is m-accretive.
(e) The operator $A_\mathfrak{a} + \lambda$ is sectorial.

It is often helpful to consider the *adjoint form* $\mathfrak{a}^*$ of a given sesquilinear form $\mathfrak{a}$ defined by

$$\mathfrak{a}^*(u, v) := \overline{\mathfrak{a}(v, u)} \qquad (u, v \in V).$$

Notice that if $\mathfrak{a}$ is continuous, then $\mathfrak{a}^*$ is so. Moreover, the form $\mathfrak{a}$ is called *symmetric*, if $\mathfrak{a} = \mathfrak{a}^*$ holds[1].

With respect to the variational operators $A_\mathfrak{a}$ and $A_{\mathfrak{a}^*}$, the following theorem is valid:

THEOREM 4.2. If $\mathfrak{a}^*$ is the adjoint form of $\mathfrak{a}$, then ${A_\mathfrak{a}}^* = A_{\mathfrak{a}^*}$. Fur-

[1] A well-known example of a symmetric form is the inner product $\langle \cdot, \cdot \rangle_H$ in a Hilbert space $H$, which differs from arbitrary symmetric sesquilinear forms only by positive definiteness.

thermore, if $\mathfrak{a}$ is symmetric, then $A_{\mathfrak{a}}$ is self-adjoint.

Given a Hilbert pair $(V, H)$, the theorem 4.2 is very helpful to construct self-adjoint linear operators in $H$ by starting with symmetric forms in $V$. This approach can especially be performed, if the form is the inner product in $V$. The operator $A$ associated with the inner product $\langle \cdot, \cdot \rangle_V$, called the *generating operator* of $(V, H)$, is coercive and even self-adjoint in $H$ by theorem 4.2.

In the following we derive the generating operator of a pair $(V, H)$ of Hilbert spaces in a straightforward way. First of all, each fixed $y \in H$ gives rise to a linear functional on $V$ according to

$$f_y(v) := \langle \iota v, y \rangle_H, \qquad (v \in V).$$

The mapping $f_y$ is also bounded, i. e. $f_y \in V_a^*$. Then, by the RIESZ representation theorem, there is $v_y \in V$ such that

$$\langle v, v_y \rangle = f_y(v) \quad \text{for all } v \in V.$$

We therefore obtain a mapping $B : H \to V$, by $By := v_y$ for $y \in H$, which proves to be linear and continuous, and the inner product in $H$ is obtained from that in $V$ by

$$\langle \iota x, y \rangle_H = \langle x, By \rangle_V \qquad (x, y \in H).$$

But the operator $B$ is also symmetric and strictly positive. These properties imply the existence of the inverse operator

$$B^{-1} : \mathfrak{D}(B^{-1}) = \mathfrak{R}(B) \subset V \to H,$$

which indeed is not necessarily bounded, but uniquely defined by the pair $(V, H)$.

The operator $B^{-1}$ is symmetric and strictly positive, too, hence there exists the square root operator $B^{-1/2}$ having the property

$$\langle B^{-1/2}x, B^{-1/2}y \rangle_H = \langle x, y \rangle_V \qquad (x, y \in H). \tag{4.3}$$

It turns out that $\mathfrak{D}(B^{-1/2}) = V$, which can be shown by contradiction. Especially for $y = x$ in (4.3) we obtain the equality

$$\|B^{-1/2}x\|_H = \|x\|_V \qquad (x \in V).$$

## 5. Sectorial Forms and Operators

In this section we introduce the notations of *closed sectorial form* and *m-sectorial operator* and show the relationships between them.

Let $H$ be a complex Hilbert space and $\mathfrak{a}:\mathfrak{D}(\mathfrak{a})\times\mathfrak{D}(\mathfrak{a})\to\mathbb{C}$ be a sesquilinear form, but defined on a linear subspace $\mathfrak{D}(\mathfrak{a})$ of $H$. Like for linear operators, the *numerical range* of $\mathfrak{a}$ is defined by

$$\Theta(\mathfrak{a}) := \{\,\mathfrak{a}(x,x) : x\in\mathfrak{D}(\mathfrak{a}),\ \|x\|=1\,\}.$$

We then denote the form $\mathfrak{a}$ to be *sectorial*, if $\Theta(\mathfrak{a})$ is contained in a sector $\Sigma_{\psi,\omega}$, that is,

$$\Theta(\mathfrak{a}) \subset \Sigma_{\psi,\omega} := \{\zeta\in\mathbb{C}:\zeta\neq\omega, arg(\zeta-\omega)<\psi\}, \tag{5.1}$$

where $\omega\in\mathbb{R}$ is the vertex and $0\le\psi<\pi/2$ the semi-angle of the sector. Notice that sectors are symmetric with respect to the real line. Especially for $\psi=0$ let $\Sigma_{0,\omega} := (\omega,\infty)$, and for $\psi=\pi/2$ we have $\Sigma_{\pi/2,0}=\mathbb{C}_+$. We also say that $A$ is $(\psi,\omega)$-sectorial, if (5.1) holds.

Now, if $\mathfrak{a}$ is symmetric and $\mathfrak{a}=\mathfrak{a}^*$, respectively, then $\Theta(\mathfrak{a})\subset\mathbb{R}$ and $\omega\le\inf\Theta(\mathfrak{a})$, so in this case we denote $\mathfrak{a}$ to be *half-bounded* or *bounded below* and call $\omega$ the lower bound of $\mathfrak{a}$. Moreover, if $\mathfrak{a}$ is half-bounded and $(\psi,\omega)$-sectorial, then the estimates

$$\begin{cases} \operatorname{Re}\mathfrak{a}(x,x) \ge \omega\,\|x\|^2 \\ \operatorname{Im}\mathfrak{a}(x,x) \le \tan\psi\cdot(\operatorname{Re}\mathfrak{a}(x,x)-\omega\,\|x\|^2) \end{cases}$$

hold for all $x\in\mathfrak{D}(\mathfrak{a})$.

Let $\operatorname{Re}\mathfrak{a} := (\mathfrak{a}+\mathfrak{a}^*)/2$ be the *real part* of $\mathfrak{a}$. Then for every $\alpha>0$ the slightly modified form $[\cdot,\cdot]_\alpha := \operatorname{Re}\mathfrak{a}-\omega+\alpha$ constitutes an inner product on the linear space $\mathfrak{D}(\mathfrak{a})$. It turns out that the norms $|\cdot|_\alpha$ derived from these inner products $[\cdot,\cdot]_\alpha$ are equivalent to each other. We especially take $\alpha=1$, pay attention to the inner product

$$[x,y]_1 = \operatorname{Re}\mathfrak{a}(x,y)-\omega\langle x,y\rangle+\langle x,y\rangle, \qquad \text{for all } x,y\in\mathfrak{D}(\mathfrak{a})$$

and distinguish between the two cases with respect to completeness of the norm

$$|x|_1 := [x,x]_1{}^{1/2} \qquad (x\in\mathfrak{D}(\mathfrak{a})).$$

In both cases it is our goal to obtain a Hilbert space $H_\mathfrak{a}$ with inner product $[x,y]_1$ and norm $|x|_1$ for $x,y\in H_\mathfrak{a}$, together with a linear embedding $\iota:H_\mathfrak{a}\to H$, which should be continuous if possible, so $(H_\mathfrak{a},H)$ hopefully becomes into a pair of Hilbert spaces.

1. The normed linear space $(\mathfrak{D}(\mathfrak{a}),|\cdot|_1)$ is *complete*.
   We denote the form $\mathfrak{a}$ *closed* in this case, and $\mathfrak{D}(\mathfrak{a})$ becomes into a Hilbert space in its own right, hence we may define $H_\mathfrak{a} := \mathfrak{D}(\mathfrak{a})$

in this case. Notice that the norm $|\cdot|_1$ on $H_\mathfrak{a}$ is stronger than the norm $\|\cdot\|$ in $H$ restricted to $H_\mathfrak{a}$, so the natural inclusion mapping $\iota : H_\mathfrak{a} \to H$ proves to be continuous.

2. The normed linear space $(\mathfrak{D}(\mathfrak{a}), |\cdot|_1)$ is *not* complete rsp. $\mathfrak{a}$ is not closed.
   We define the Hilbert space $H_\mathfrak{a}$ as the completion of $\mathfrak{D}(\mathfrak{a})$ with respect to the norm $|\cdot|_1$. Moreover, Cauchy sequences in $H_\mathfrak{a}$ are also Cauchy sequences in $H$, hence for every $x \in H_\mathfrak{a}$ there is $x' \in H$ such that $x' = \iota x$ for a linear map $\iota : H_\mathfrak{a} \to H$.
   We want to extend the form $\mathfrak{a}$ from $\mathfrak{D}(\mathfrak{a})$ to $H_\mathfrak{a}$ as well, and this will be done by defining
   $$\overline{\mathfrak{a}}(x, y) := \lim_{n \to \infty} \mathfrak{a}(x_n, y_n)$$
   for $x, y \in H_\mathfrak{a}$ and arbitrary sequences $\{x_n\}_{n\in\mathbb{N}}, \{y_n\}_{n\in\mathbb{N}} \subset \mathfrak{D}(\mathfrak{a})$ with $x_n \to x$ and $y_n \to y$. It turns out that the definition of $\overline{\mathfrak{a}}(x, y)$ is independent of the sequences $\{x_n\}_{n\in\mathbb{N}}$ and $\{y_n\}_{n\in\mathbb{N}}$, so this value is in fact well-defined for $x, y \in H_\mathfrak{a}$.

In summary, we obtain in both cases a Hilbert space $H_\mathfrak{a}$ with inner product $[\cdot,\cdot]_1$ and norm $|\cdot|_1$ together with a linear embedding $\iota : H_\mathfrak{a} \to H$, but which in the second case is not necessarily continuous.

Regarding the second case, the form $\mathfrak{a}$ is said to be *closable*, if the mapping $\iota$ is continuous. This definition especially means that every closed form is closable. If $\mathfrak{a}$ is closable, then the final result in both cases will be that there exists a closed form, namely $\mathfrak{a}$ itself in the first and $\overline{\mathfrak{a}}$ in the second case, together with a continuous embedding of $H_\mathfrak{a}$ into $H$.

Let us continue with the concept of *sectorialness* of a closed linear operator in a complex Banach or Hilbert space. This class of operators has been introduced and first studied by T. KATO. Let again $\Sigma_{\psi,\omega}$ be an open sector with vertex $\omega \in \mathbb{R}$ and semi-angle $0 < \psi \leq \pi$. In his book [1], KATO denotes a linear operator $A$ in a Hilbert space to be *sectorial*, if the following conditions are fulfilled:

(S-1) $A$ is quasi-accretive, i. e. $\langle (A + \omega)x, x\rangle \geq 0$ for some $\omega \in \mathbb{R}$ and $x \in \mathfrak{D}(A)$,

(S-2) the numerical range $\Theta(A)$ is a subset of the sector $\Sigma_{\psi,\omega}$.

Furthermore, $A$ is called *m-sectorial*, if it is at once sectorial and m-quasi-accretive, that is, $A + \omega$ is m-accretive for some $\omega \in \mathbb{R}$.

We finally want to elaborate the correspondence between closed sectorial forms and m-sectorial operators. It should be clear that if $A$ is a m-sectorial operator in $H$ with domain of definition $\mathfrak{D}(A)$, then the sesquilinear form

$$\mathfrak{a}_A : \begin{cases} \mathfrak{D}(\mathfrak{a}_A) := \mathfrak{D}(A), \\ \mathfrak{a}_A(x,y) := \langle Ax, y\rangle \text{ for } x, y \in \mathfrak{D}(\mathfrak{a}_A) \end{cases}$$

is sectorial and closed.

Reversely, let $\mathfrak{a}_0$ be a closable sectorial form in $H$ with domain $\mathfrak{D}(\mathfrak{a})$. We consider the closed completion $\mathfrak{a}$ on the space $H_\mathfrak{a}$. A densely defined linear subspace $D \subset \mathfrak{D}(\mathfrak{a})$ is denoted a *core* of $\mathfrak{a}$, or equivalently, the set $D$ is a core of $\mathfrak{a}$, if $\mathfrak{a}$ is the closure of the restriction of $\mathfrak{a}$ to $D$.

KATO*'s first representation theorem* claims that a densely defined and closed sectorial form gives rise to a m-sectorial linear operator:

THEOREM 5.1. Let $\mathfrak{a}$ be a densely defined and closed sectorial form in $H$ with vertex $\omega$, semi-angle $\psi$ and domain of definition $H_\mathfrak{a}$, and let $\iota$ be the continuous embedding from $H_\mathfrak{a}$ into $H$. Then there is a m-sectorial operator $A_\mathfrak{a}$ in $H$, whose domain of definition $\mathfrak{D}(A_\mathfrak{a})$ is defined as follows: Any element $x \in H$ belongs to $\mathfrak{D}(A_\mathfrak{a})$, if $x = \iota x_0$ for some $x_0 \in H_\mathfrak{a}$ and $\mathfrak{a}(x_0, z) = \langle y, \iota z\rangle$ holds for some $y \in H$ and all $z \in H_\mathfrak{a}$, which suggests to define $A_\mathfrak{a} x := y$.

The operator $A_\mathfrak{a}$ owns a couple of properties:

(a) $\mathfrak{D}(A_\mathfrak{a})$ is dense in $H$ and the preimage $\iota^{-1}\mathfrak{D}(A_\mathfrak{a})$ is dense in $H_\mathfrak{a}$.
(b) If $x \in H_\mathfrak{a}$, $y \in H$ and $\mathfrak{a}(x,y) = \langle \iota x, y\rangle$ for all $x$ in a core of $\mathfrak{a}$, then $\iota x \in \mathfrak{D}(A_\mathfrak{a})$ and $y = A(\iota x)$.
(c) $\Theta(A_\mathfrak{a})$ is a dense subset of $\Theta(\mathfrak{a})$.
(d) The adjoint form $\mathfrak{a}^*$ is also closed and sectorial with vertex $\omega$ and semi-angle $\psi$.
(e) The adjoint $A_\mathfrak{a}{}^*$ of $A_\mathfrak{a}$ is the *m*-sectorial operator associated with the adjoint form $\mathfrak{a}^*$.

Thus m-sectorial linear operators are in one-to-one correspondence with closed sectorial forms.

## 6. Self-Adjoint Extensions of Half-Bounded Operators

We want to apply the results of the last section to the following class of operators: a *symmetric* linear operator $A$ is called *half-bounded*,

if there is $\theta \in \mathbb{R}$ such that

$$\langle A\,x, x\rangle \;\geq\; \theta\,\|x\|^2 \qquad \text{for all } x \in \mathfrak{D}(A),$$

or equivalently, if $\Theta(A) \subset [\theta, \infty)$ for some $\theta \in \mathbb{R}$. The operator $A$ is called *positive* (*strictly positive*), if $\theta = 0$ ($\theta > 0$) can be chosen. In case of $\theta > 0$, the Cauchy-Schwarz inequality implies that $A$ is also bounded below, hence $A$ is surely injective. In addition, if $A$ is closed, then the range $\mathfrak{R}(A)$ is closed in $H$, which follows from the bounded-below theorem, but $\mathfrak{R}(A)$ may be a proper subspace of $H$.

The main result in this section is given by the fact that each densely defined and half-bounded linear operator $A$ acting in a Hilbert space owns at least one *self-adjoint extension.*

First notice that if $A$ is half-bounded with $\theta \in \mathbb{R}$, then it is $(0, \theta)$-sectorial, and the sesquilinear form $\mathfrak{a}(x, y) := \langle Ax, y\rangle$ is sectorial and densely defined. But $\mathfrak{a}$ is also closable, so we may switch to the sesquilinear form $\bar{\mathfrak{a}}$ defined on the Hilbert space $H_{\mathfrak{a}}$, called the *energetic space* of the operator $A$. Then by theorem 5.1 there is a m-sectorial and even self-adjoint linear operator $A_{\mathfrak{a}}$, that forms an extension of the half-bounded operator $A$, called the *Friedrichs extension* of $A$.

The domain of definition $\mathfrak{D}(A_{\mathfrak{a}})$ of the operator $A_{\mathfrak{a}}$ coincides with the linear subspace $H_{\mathfrak{a}} \cap \mathfrak{D}(A^*)$. Moreover, the operator $A_{\mathfrak{a}}$ owns a useful property: if one adds to the half-bounded operator $A$ the multiplication operator $x \in H \mapsto \theta x$ with $\theta \in \mathbb{R}$, then the Friedrichs extensions of $A$ and $A + \theta$ are related to each other by $(A + \theta)_{\mathfrak{a}} = A_{\mathfrak{a}} + \theta$.

## 7. Variational Evolution Equations

Having the concept of Hilbert-Sobolev space available, we want to review the variational approach to first and second order evolution equations, which has been developed by J.-L. LIONS in the midth of the last century. It can be seen as the straightforward extension of the LAX-MILGRAM theory from stationary equations to evolutional ones.

Let $\tau > 0$ and denote $\lambda$ the Lebesgue-Borel measure on $[0, \tau)$. By an *abstract variational problem of parabolic type* in a real Hilbert space triple $(V, H, V^*)$ we mean a $\lambda$-nearly everywhere in $(0, \tau)$ valid linear operator-differential equation

$$\begin{cases} u'(t) + \mathcal{A}_{\mathfrak{a}}(t)\, u(t) = f(t), \\ u(0) = u_0, \end{cases} \tag{7.1}$$

where $u_0 \in H$, $f : [0,\tau) \to V^*$ and $\mathcal{A}_{\mathfrak{a}t\in[0,\tau)}$ is a family of linear operators $\mathcal{A}_{\mathfrak{a}} : V \to V^*$.

Let $\mathfrak{a}(t)$ be the bilinear form associated with the operator $\mathcal{A}_{\mathfrak{a}}(t)$. Then we call $u \in H^1(0,\tau;\, V,\, V^*)$ a *weak solution* of (7.1), if $u(0) = u_0$ and

$$\langle u'(t)\,, v(t)\rangle + \mathfrak{a}(t, u(t), v(t)) = f(t)(v)$$

for all $v \in V$ and $\lambda$-nearly everywhere $t \in [0,\tau)$.

Regarding the existence and uniqueness of a *weak solution* for the initial value problem (7.1), the following theorem is of fundamental meaning:

THEOREM 7.1. Let $\{\mathfrak{a}(t)\}_{t\in[0,\tau)}$ fulfill the following conditions:

(a) the mapping $t \in [0,\tau) \mapsto \mathfrak{a}(t,u,v)$ is Lebesgue-measurable for every fixed pair $(u,v) \in V \times V$,

(b) $\mathfrak{a}$ is *uniformly* bounded in $[0,\tau)$, that is, there is a constant $M \geq 0$ such that

$$|\mathfrak{a}(t,u,v)| \leq M \cdot \|u\|_V \, \|v\|_V \qquad \text{for all } u,v \in V,\ t \in [0,\tau)\,,$$

(c) $\mathfrak{a}(t)$ is quasi-coercive for each $t \in [0,\tau)$.

Then for each $u_0 \in H$ and each $f \in L^2(0,\tau;\, V^*)$ there is exactly one solution $u \in H^1(0,\tau;\, V,\, V^*)$ of (7.1), which continuously depends on the data $u_0$ and $f$.

The proof of this theorem has been published in [2] as *Théorème 1.1*, page 46. Both terms $u'$ and $\mathcal{A}_{\mathfrak{a}}(t)u$ in (7.1) belong to the same space $L^2(0,\tau;\, V^*)$ as $f$, which gives rise to say that the problem has *maximal regularity* in $V^*$.

The concept of weak solution makes also sense for linear evolution equations of second order.

An *abstract hyperbolic variational problem* with respect to the Hilbert space triple $(V, H, V^*)$ and a family of bilinear forms $\mathfrak{a}(t) : [0,\tau] \times V \times V \to \mathbb{R}$ is given by

$$\begin{cases} \langle u''(t)\,, v(t)\rangle + \mathfrak{a}(t, u(t), v(t)) = \langle f(t), v\rangle \quad (v \in V, \\ u(0) = u_0 \in V, \\ u'(0) = u_1 \in H, \end{cases} \tag{7.2}$$

where $\langle \cdot, \cdot \rangle$ is the inner product in $H$.

The next theorem shows that problem 7.2 is *weakly solvable* under a couple of conditions on the involved bilinear forms:

THEOREM 7.2. Let $\{\mathfrak{a}(t)\}_{t\in[0,\tau)}$ be a family of bilinear forms on $V$ such that

(a) $t \mapsto \mathfrak{a}(t)$ is *uniformly* bounded in $[0,\tau)$,
(b) $\mathfrak{a}(t)$ is $H$-elliptic for each $t \in [0,\tau)$,
(c) $\mathfrak{a}(t)$ satisfies the symmetry condition $\mathfrak{a}(t,u,v) = \mathfrak{a}(t,v,u)$ for all $u, v \in V$, $t \in [0,\tau)$,
(d) the mapping $t \mapsto \mathfrak{a}(t,u,v)$ is continuously differentiable in $[0,\tau)$ for each fixed $(u,v) \in V \times V$,
(e) there is $M_1 \geq 0$ such that

$$|\frac{d}{dt}\mathfrak{a}(t,u,v)| \;\leq\; M_1 \cdot \|u\|_V \, \|v\|_V \qquad \text{for all } \; t \in [0,\tau).$$

Then the initial value problem (7.2) has a unique weak solution $u \in H^1(0,\tau;\, V,\, V^*)$ for all $u_0 \in V$, $u_1 \in H$ and $f \in L^2(0,\tau;H)$.

The symmetry condition (c) in 7.2 is essential to prove this theorem. Moreover, the solution map

$$R : L^2(0,\tau;H) \times V \times H \to L^2(0,\tau;H)$$

given by $(f, u_0, u_1) \mapsto u$ is linear in each argument and continuous, thus problem (7.2) proves to be well-posed.

## 8. Applications

In this final section we want to illustrate the usage of abstract sesquilinear forms in Hilbert space for the analysis of linear equations of the form $\mathcal{A}u = f$, where $\mathcal{A}$ is a linear partial differential operator and $u, f$ are real or complex-valued mappings defined on an open set $G$ within an Euclidean space $\mathbb{R}^d$.

A differential expression $\mathcal{A}$ of order $2m$ is said to be in *divergence form*, if there are functions $a_{\alpha\beta} \in C^{|\alpha|}(G)$ for $|\alpha|, |\beta| \leq m$ such that $\mathcal{A}$ can be written as

$$\mathcal{A} \;:=\; \sum_{|\alpha|\leq m, |\beta|\leq m} (-1)^{|\alpha|}\, \partial^\alpha (\, a_{\alpha\beta}(x)\, D^\beta). \tag{8.1}$$

In opposite to this definition we call an arbitary differential expression of order $2m$ given by

$$\mathcal{A}'(x,\,\cdot\,) := \sum_{|\alpha|\le 2m} c_\alpha(x)\,\partial^\alpha \tag{8.2}$$

to be *in non-divergence form.*

It is a matter of fact that every linear differential expression $\mathcal{A}$ in divergence form can be transformed into an expression $\mathcal{A}'$ in *non-divergence form.* Reversely, a non-divergence form expression $\mathcal{A}'$ given by (8.2) can be transformed into an expression $\mathcal{A}$ in divergence form (8.1), if $c_\alpha \in C^{2m-|\alpha|}(G)$ for $|\alpha| \le 2m$, that is, if the coefficients are sufficiently many differentiable in $G$. By comparing the expressions (8.2) and (8.1), it turns out that $\mathcal{A}'$ is strongly elliptic in $G$ if and only if $\mathcal{A}$ is so.

Next we elaborate the purpose of a differential operator being in divergence form. Let $C_c^\infty(G)$ be the space of *test functions* and furnished with the inner product

$$\langle \phi, \psi \rangle := \int_G \phi\,\overline{\psi}\,dx \quad \text{for all } \phi, \psi \in C_c^\infty(G).$$

Now, applying $\mathcal{A}$ given in divergence form to $\phi \in C_c^\infty(G)$, then multiplying $\mathcal{A}\phi$ with $\overline{\psi} \in C_c^\infty(G)$ and finally performing $m$-times partial integration leads to

$$B_{\mathcal{A}}(\phi,\psi) := \langle \mathcal{A}\phi, \psi \rangle = \sum_{|\alpha|\le m, |\beta|\le m} \int_G a_{\alpha\beta}\,\partial^\alpha\phi\,D^\beta\overline{\psi}\,dx, \tag{8.3}$$

since all the boundary terms vanish. The mapping

$$B_{\mathcal{A}} := C_c^\infty(G) \times C_c^\infty(G) \to \mathbb{C}$$

is a sesquilinear form, called the *Dirichlet form* of $\mathcal{A}$ in $G$. If the coefficent functions $a_{\alpha\beta}$ of $\mathcal{A}$ are measurable and essentially bounded, then the mapping $B_{\mathcal{A}}$ is continuous and thus can continuously be extended to the product space $H_0^m(G) \times H_0^m(G)$, where $H_0^m(G)$ is the Sobolev space with null boundary conditions, that is, the completion of $C_c^\infty(G)$ with respect to the Sobolev norm $|\cdot|_m$.

We are mainly interested in a $2m$-order linear differential operator $\mathcal{A}$ given in divergence form, that together with the underlying domain $G$ fulfills the following conditions:

(I) $G$ is bounded in $\mathbb{R}^d$,

(I') $G$ satisfies the $m$-th *extension property*, that is, the linear space $R^0(G)$ consisting of those functions $\phi \in W^{m,p}(G)$, which are restrictions of functions $\phi_{ext} \in C_c^\infty(\mathbb{R}^d)$ to $G$, is

dense in $W^{m,p}(G)$.

(II) $\mathcal{A}$ is uniformly strongly elliptic in $G$, that is, there is $\theta > 0$ such that $\operatorname{Re}(-1)^m p(x,\xi) \geq \theta \cdot |\xi|^{2m}$ for all $\xi \neq 0$ and $x \in \overline{G}$.

(III) the coefficient functions $a_{\alpha\beta}$ are essentially bounded in $G$, that is, $a_{\alpha\beta} \in L^\infty(G)$,

(III') the coefficient functions $a_{\alpha\beta}$ of highest order $|\alpha| + |\beta| = 2m$ are uniformly continuous in $\overline{G}$.

GÅRDING*'s inequality* is the main assertion in this section and can be seen as a bridge from the theory of linear-elliptic differential operators to abstract functional analysis:

THEOREM 8.1. Let the domain $G$ fulfill the conditions (I), (I') and let the operator $\mathcal{A}$ of order $2m$ fulfill the conditions (II), (III), (III'). Then there is $\lambda_0 \in \mathbb{R}$ and $a_0 > 0$ such that for all $u \in H_0^m(G)$

$$B_{\mathcal{A}}(u,u) + \lambda_0 \|u\|^2 = \langle (\mathcal{A} + \lambda_0) u, u \rangle \geq a_0 |u|_m{}^2. \tag{8.4}$$

In other words, GÅRDING's inequality quotes that under the mentioned conditions the Dirichlet form associated with the operator $\mathcal{A}+\lambda_0$ is coercive, hence the LAX-MILGRAM theorem may be applied.

The most important class of linear partial differential operators in divergence form with respect to applications is that of *second order*, that is, our operator $\mathcal{A}$ is of the form

$$\mathcal{A}(x,\cdot) = -\sum_{i=1}^{d} \partial_i \Big(\sum_{j=1}^{d} a_{ij}(x)\Big) + \sum_{j=1}^{d} (b_j(x)\,\partial_j) + c(x), \tag{8.5}$$

which is nothing but (8.1) for $m = 1$.

The Dirichlet form $B_{\mathcal{A}}$ belonging to $\mathcal{A}$ given by (8.5) is the sesquilinear form defined by

$$B_{\mathcal{A}}(u,v) = \int_G \Big(\sum_{i,j=1}^{d} a_{ij}\,\partial_i u\,\partial_j \overline{v} + \sum_{j=1}^{d} b_j\,\partial_j u\,\overline{v} + c\,u\,\overline{v}\Big)\,dx. \tag{8.6}$$

In this case, GÅRDING's inequality remains even true if condition (I)' and (III') are dropped:

THEOREM 8.2. Let $G \subset \mathbb{R}^d$ fulfill (I) and the differential operator $\mathcal{A}$ of second order and given in divergence form fulfill (II) and (III). Then there is $\lambda_0 \in \mathbb{R}$ and $a_0 > 0$ such that

$$B_{\mathcal{A}}(u,u) \,+\, \lambda_0 \,\|u\|^2 \,=\, \langle\, (\mathcal{A}+\lambda_0)\, u, u\rangle \,\geq\, a_0\, |u|_1{}^2$$

for all $u \in H_0^1(G)$, where $a_0$ is the uniform lower bound of the eigenvalues of all the matrices $A(x)$ with $x \in G$.

To prove GÅRDING's inequality in the special case of second order operators, first recall YOUNG*'s inequality* $ab \leq \epsilon a^2 + 4b^2/\epsilon$ for $a, b \in \mathbb{R}$ and any $\epsilon > 0$. By the strict ellipticity condition for $\mathcal{A}$ and with $b_0 := \max_{1\leq j\leq d} \|b_j\|_\infty$ and $c_0 := \|c\|_\infty$ we have

$$\begin{aligned} B_{\mathcal{A}}(u,u) \,&=\, \int_G \Big(\sum_{i,j=1}^{d} a_{ij}\, \partial_i u\, \partial_j \overline{u} \,+\, \sum_{j=1}^{d} b_j\, \partial_j u\, \overline{u} \,+\, c\, u\, \overline{u}\Big)\, dx \\ &\geq\, \theta \int_G |\nabla u|^2\, dx \,-\, b_0 \int_G |\nabla u|\, |u|\, dx \,-\, c_0\, \|u\|^2 \\ &\geq\, \theta \int_G |\nabla u|^2\, dx \,-\, b_0 \epsilon \int_G |\nabla u|^2\, dx \,-\, \frac{b_0}{4\epsilon} \int_G |u|^2\, dx \,-\, c_0\, \|u\|^2 \end{aligned}$$

for some $\epsilon > 0$. Let us chose $\epsilon$ small enough to ensure $b_0\, \epsilon \leq \frac{\theta}{2}$, then

$$\begin{aligned} B_{\mathcal{A}}(u,u) \,&\geq\, \frac{\theta}{2} \int_G |\nabla u|^2\, dx \,-\, \Big(\frac{b_0}{4\epsilon} + c_0\Big)\, \|u\|^2 \,= \\ &\frac{\theta}{2} \Big(\int_G \big(|u|^2 + |\nabla u|^2\big)\, dx \,-\, \Big(\frac{b_0}{4\epsilon} + c_0 + \frac{\theta}{2}\Big)\, \|u\|^2\,, \end{aligned}$$

hence with $a_0 := \theta/2$ and $\lambda_0 := b_0/4\epsilon + c_0 + \theta/2$ we obtain (8.4).

Recall again that a sesquilinear form $\mathfrak{a}$ acting in a complex Hilbert space $V$ is coercive, if $\operatorname{Re} \mathfrak{a}(x,x) \geq \theta\, \|x\|^2$ holds for all $x \in V$. In the following we again consider the Dirichlet form $B_{\mathcal{A}}$ of a second-order differential operator $\mathcal{A}$ given in divergence form and by (8.5). A short computation shows that $B_{\mathcal{A}}$ is coercive, if the function $c$ is large enough in comparison with the sum of partial derivatives of the functions $b_k$. More precisely, $B_{\mathcal{A}}$ proves to be coercive if

$$c(x) \,\geq\, \frac{1}{2} \sum_{k=1}^{d} \frac{\partial b_k}{\partial x_k}(x) \qquad (x \in G). \tag{8.7}$$

On the other hand, if the matrix $A(x) = (a_{ij}(x))_{i,j=1}^d$ consistent of the second-order coefficients of $\mathcal{A}$ fails to be positive definite for all $x \in G'$, where $G' \subset G$ has positive measure, then $B_{\mathcal{A}}$ cannot be coercive on $H_0^1(G)$. Otherwise, if $A(x)$ is positive definite in all $x \in G$ and condition (8.7) does not hold. then the functions

$b_j$ disturb the coercivity of the form $B_{\mathcal{A}}$. But after all, the form $B_{\mathcal{A}}$ is at least quasi-coercive, which is nothing but the assertion of GÅRDING's inequality.

Let us especially consider the Dirichlet form $B_{-\Delta}$ associated with the negative Laplacian $-\Delta$, which adopts the simple form

$$B_{-\Delta}(u,v) \;=\; \int_G \nabla u \cdot \nabla \overline{v}\, dx \tag{8.8}$$

for $u, v \in H_0^1(G)$. GÅRDING's inequality for this form reads as

$$B_{-\Delta}(u,u) \;+\; \|u\|^2 \;=\; \int_G (|\nabla u|^2 + |u|^2)\, dx \;=\; |u|_1{}^2,$$

so $a_0 = \lambda_0 = 1$ can be chosen in this case.

The assertion that $B_{-\Delta}$ is coercive in bounded domains, is of course a consequence of GÅRDING's inequality, but may also proved directly by POINCARÉ*'s inequality* even for non-bounded domains. Let the mapping $|\cdot|_e : H_0^1(G) \to [0,\infty)$ be defined by

$$|u|_e \;:=\; \left( \int_G \nabla u\, \nabla \overline{u}\; dx \right)^{1/2},$$

which fulfills all the properties of a norm. It is denoted the *energetic norm* in $H_0^1(G)$. Suppose that $G$ has at least a bound in one direction, say $|x_1| \le \delta$ for all $x = (x_1, \ldots, x_d) \in G$ and some $\delta > 0$. Then there is a constant $C(G) > 0$ such that

$$|u|_1 \;\le\; C(G) \cdot |u|_e \tag{8.9}$$

for all $u \in H_0^1(G)$, where the lower bound $C(G) > 0$ only depends on the shape of $G$. This inequality claims that in $H_0^1(G)$ the energetic norm $|\cdot|_e$ and the Sobolev norm $|\cdot|_1$ are equivalent to each other.

POINCARÉ's *inequality* also implies that the form $B_{-\Delta}$ as defined in (8.8) is coercive. The linear operator $A_{\mathfrak{a}}$ associated with $\mathfrak{a} = B_{-\Delta}$ has domain of definition

$$\mathfrak{D}(A_{\mathfrak{a}}) \;=\; \{\, u \in H_0^1(G) : A_{\mathfrak{a}} u \in L^2(G) \,\}$$

with values $A_{\mathfrak{a}} u := -\Delta u$ in the sense of weak derivatives. This operator is in fact the realization of the negative Laplacian in Hilbert space $L^2(G)$ and under homogeneous Dirichlet conditions. By theorem 4.1, $A_{\mathfrak{a}}$ is a densely defined and closed operator.

Let us now turn back to higher order differential operators and formulate the existence and uniqueness assertion for stationary

equations of the form $\mathcal{A}u = f$, where

(DP-1) the differential operator is of order $2m$, strongly elliptic in a *bounded* domain $G \subset \mathbb{R}^d$ and given in divergence form, hence

$$\mathcal{A} := \sum_{|\alpha|\le m,|\beta|\le m} (-1)^{|\alpha|}\, \partial^\alpha (\, a_{\alpha\beta}(x)\, D^\beta),$$

(DP-2) the coefficients fulfill $a_{\alpha\beta} \in L^\infty(G)$ and the top-order coefficients are uniformly continuous in $\overline{G}$.

Our aim is to solve the *homogeneous Dirichlet problem*

$$\begin{cases} (\mathcal{A}u)(x) = f(x) & (x \in G), \\ u(x) = \partial u/\partial\nu(x) = \ldots = \partial u^{m-1}/\partial\nu^{m-1}(x) = 0 & (x \in \partial G). \end{cases}$$

Let $V = H_0^m(G)$ and $H = L^2(G)$, so the pair $(V, H)$ forms a Hilbert triple. By GÅRDING's inequality (8.4), the Dirichlet form $B_{\mathcal{A}}(u,v) := \langle \mathcal{A}u, v\rangle$ for $u, v \in V$ of $\mathcal{A}$ is quasi-coercive with some $\lambda_0 \in \mathbb{R}$. Then the LAX-MILGRAM theorem implies the *existence and uniqueness theorem* for the homogeneous Dirichlet problem:

THEOREM 8.3. For $\lambda \ge \lambda_0$, the variational equation

$$B_{\mathcal{A}}(u,v) + \lambda\langle u, v\rangle = f(v) \quad (v \in H_0^m(G)) \tag{8.10}$$

owns for each $f \in H^{-m}(G)$ a unique solution $u \in H_0^m(G)$.

The function $u$ is called the *weak solution* of the homogeneous Dirichlet problem.

Having theorem 8.3 available, it is the task of *elliptic regularity theory* to show, that if $G$ is bounded, $\partial G$ is sufficiently smooth and $f \in H^k(G)$ for some $k \in \mathbb{N}_0$, then the solution $u$ of (8.10) actually belongs to the space $H^{k+2m}(G) \cap H_0^m(G)$.

Moreover, let us assume that the existence of a weak solution for equation $\mathcal{A}u = f$ under homogeneous Dirichlet boundary conditions is ensured. Then the question arises, under which conditions the weak solution $u \in H_0^m(G)$ is classical, that is, $u$ is contained in the space $C^{2m}(G) \cap C^m(\overline{G})$.

In the following we present only few results on regularity theory and restrict ourselves to the second order differential operator

$$\mathcal{A}_2(x,u) \;=\; -\sum_{i,j=1}^{d} \partial_i(a_{ij}(x)\,\partial_j u(x)) + \sum_{j=1}^{d} b_j(x)\,\partial_j u(x) + c(x)\,u(x).$$

Basically, we distinguish between *inner regularity* and *regularity up to the boundary.*

Starting with *inner regularity*, let $\mathcal{A}_2$ be strongly elliptic and of second order, whose coefficients fulfill $a_{ij} \in C^1(G)$, $b_i \in L^\infty(G)$ and $c \in L^\infty(G)$. If $f \in L^2(G)$ and $u \in H^1(G)$ is a weak solution of $\mathcal{A}_2\, u = f$, then the following propositions are valid:

1. $u \in H^2_{loc}(G)$, that is, for each open subset $G' \Subset G$[2] the function $u\,|_{G'}$ belongs to the space $H^m(G')$,
2. for each open subset $G' \Subset G$ the estimate

$$\|u\|_{H^2(G')} \;\le\; C \cdot [\,\|f\|_{L^2(G)} + \|u\|_{L^2(G)}\,]$$

holds, where the constant $C$ only depends on $G'$, $G$ and the coefficients of the operator $\mathcal{A}_2$.

Notice that there are no assumptions on smoothness of the boundary $\partial G$ required.

A similar result is obtained, called *higher inner regularity*, if one requires higher smoothness properties for the function $f$ and the coefficients of $\mathcal{A}_2$. Let $a_{ij}, b_i, c \in C^{m+1}(G)$, $f \in H^m(G)$ for some $m \in \mathbb{N}$. If $u \in H^1(G)$ is a weak solution of the equation $\mathcal{A}_2 u = f$, then we have $u \in H^{m+2}_{loc}(G)$ and

$$\|u\|_{H^{m+2}(G')} \;\le\; C \cdot [\,\|f\|_{H^m(G)} + \|u\|_{L^2(G)}\,],$$

where $C$ again only depends on the sets $G'$, $G$ and the coefficients of $\mathcal{A}_2$. Finally, if $u \in H^1(G)$ is the weak solution of $\mathcal{A}_2\, u = f$ and $a_{ij}, b_i, c, f \in C^\infty(G)$, then we even have $u \in C^\infty(G)$.

We now turn to *regularity up to the boundary.* This notation corresponds to assertions about the membership of the weak solution to one of the Sobolev spaces $H^m(G)$. However, to obtain this kind of regularity, one has to postulate sharper requirements on the coefficients of the differential operator and certain smoothness properties of the boundary of $G$.

This kind of regularity means that if $k \in \mathbb{N}_0$, $a_{ij}, b_j, c \in C^k(\overline{G})$, $f \in H^k(G)$, the boundary $\partial G$ is of class $C^{k+2}$ and $u \in H^1_0(G)$

[2] $G' \Subset G$ means that $G'$ is *compactly embedded* in $G$, i. e. there is a compact set $K$ such that $G' \subset K \subset G$.

is a weak solution of the homogeneous Dirichlet problem, then $u \in H^{k+2}(G)$ and the following Schauder estimate is valid:

$$\|u\|_{H^{k+2}(G)} \leq C \cdot [\, \|f\|_{H^k(G)} + \|u\|_{L^2(G)} \,].$$

There are statements about higher regularity of $u$ as well: let $a_{ij}, b_i, c \in C^\infty(\overline{G})$, $f \in C^\infty(G) \cap L^2(G)$ and $\partial G$ be of class $C^\infty$, then each solution $u \in H^1(G)$ of the homogeneous Dirichlet problem is indefinitely differentiable (WEYL*'s lemma*).

Besides the Dirichlet boundary value problem, the *Neumann problem* is the second important boundary value problem for elliptic differential operators of second order and especially for the negative Laplacian, which in the following we shall sketch in the setting of Hilbert-Sobolev spaces.

Let $G \subset \mathbb{R}^d$ be a bounded domain with $C^1$-boundary, so the outer normal vector $n(x)$ is defined in every point $x \in \partial G$. Then the *homogeneous* Neumann problem reads as follows: find a function $u \in C^2(G) \cap C^1(\overline{G})$ such that

$$\begin{cases} (-\Delta u)x = f(x), & x \in G, \\ u_\nu(x) := (\nabla u)(x) \cdot n(x) = 0, & x \in \partial G \end{cases} \tag{8.11}$$

for given $f \in C(G)$. Notice that with each solution $u$ of this problem the function $u_c := u + c$ ($c \in \mathbb{R}$) is a solution, too.

The *divergence theorem* quotes that

$$\int_G \Delta u \, dx = \int_{\partial G} \nabla u \cdot n \, d\sigma$$

holds for every function $u \in C^2(G) \cap C(\partial G)$, which implies that a necessary condition for the existence of a solution of (8.11) is the validness of the equation

$$\int_G f dx = -\int_G \Delta u \, dx. \tag{8.12}$$

Conversely, if this compatiblity condition does not hold, there cannot exist a solution of problem (8.11).

The variational approach to the Neumann problem consists of multiplying each side of $-\Delta u = f$ with functions $v \in H^1(G)$ and performing partial integration on the left hand side, which leads to the variational equation

$$\int_G \nabla u \cdot \nabla v \, dx \;=\; \int_G f \, v \, dx.$$

Let $\gamma_0$ be the *trace operator* on the Sobolev space $H^1(G)$ into the space $L^2(\partial G)$. Then $u \in T_g(G) := \{\, \phi \in H^1(G) : \gamma_0 \phi = 0 \,\}$, the *trial space* for the Neumann problem, is called a *weak solution* of (8.11), if for all $v \in H^1(G)$

$$\int_G \nabla u \cdot \nabla v \, dx \;=\; \int_G f \, v \, dx.$$

If a linear differential operator $\mathcal{A}$ is given in divergence form and its Dirichlet form is coercive, then the variational approach to the heat equation associated with $\mathcal{A}$ may be applied. Consider for $f \in L^2([0,\tau), H^{-1}(G))$ and $u_0 \in L^2(G)$ the *parabolic* initial-value problem

$$\begin{cases} \langle u_t, v\rangle \;+\; B_{\mathcal{A}}[u,v] \;=\; \langle f, v\rangle \quad \text{for all } v \in H_0^1(G), \\ u(t,\cdot) \in H_0^1(G), \\ u(0) = u_0. \end{cases}$$

This problem owns a uniquely defined weak solution in the space $H^1(0,\tau; H_0^1(G), H^{-1}(G))$, which fulfills the variational equation $\lambda$-nearly everywhere in $[0,\tau)$. Notice that this assertion is especially true if $\mathcal{A} = -\Delta$.

Again, if $\mathcal{A}$ is given in divergence form, the *hyperbolic* initial-value problem in variational form is given by

$$\begin{cases} \langle u_{tt}, v\rangle \;+\; B_{\mathcal{A}}[u,v] \;=\; \langle f, v\rangle \quad \text{for all } v \in H_0^1(G), \\ u(t,\cdot) \in H_0^1(G), \\ u(0) = u_0, \\ u_t(0) = u_1, \end{cases}$$

where the coefficients $a_{ij}$ of the operator $\mathcal{A}$ must fulfill the symmetry conditions

$$a_{ij}(x) \;=\; \overline{a_{ji}(x)} \quad 1 \le i,j \le d, \;\; x \in G.$$

If $f$ belongs to $L^2([0,\tau), L^2(G))$, $u_0 \in H_0^1(G)$ and $u_1 \in L^2(G)$, then this problem has exactly one weak solution $u \in L^2(0,\tau; H_0^1(G))$ such that $u(0) = u_0$, $u_t(0) = u_1$ and $u(t) \in H_0^1(G)$ for $t \in [0,\tau)$ $\lambda$-nearly everywhere.

Since the Dirichlet form of the expression $-\Delta$ is symmetric, the variational initial-value problem for the classical wave equa-

tion $u'' - \Delta u = f$ owns a uniquely defined weak solution $u \in H^1(0,\tau; H_0^1(G), H^{-1}(G))$, if the functions $f, u_0, u_1$ fulfills the mentioned prerequisites.

## References

[1] T. Kato: *Perturbation Theory for Linear Operators*, Springer Verlag, Grundlehren der mathematischen Wissenschaften, Band 132, 1995.

[2] J.-L. Lions: *Équations Différentielles Opérationelles et Problèmes aux Limites*. Springer, Grundlehren der mathematischen Wissenschaften, Band 111, 1961.

[3] M. Renardy, R. Rogers: *An Introduction to Partial Differential Equations*. Texts in Applied Mathematics 13, Springer, 2004.

[4] R. Showalter: *Hilbert Space Methods in Partial Differential Equations*, Dover Publications, 2010.